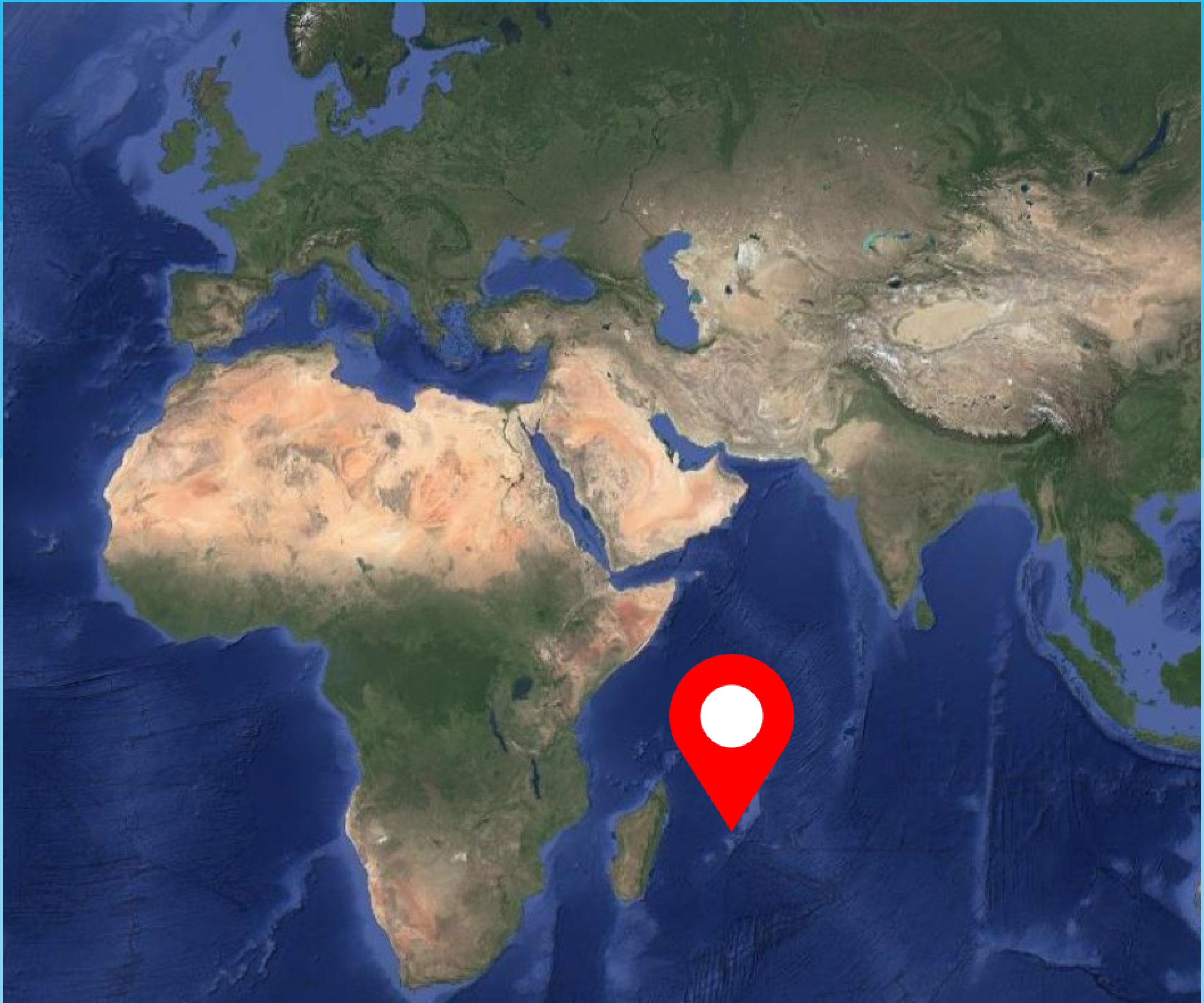
How (And Why) To Build a Cloud Service Provider From Scratch

By Alex Bissessur

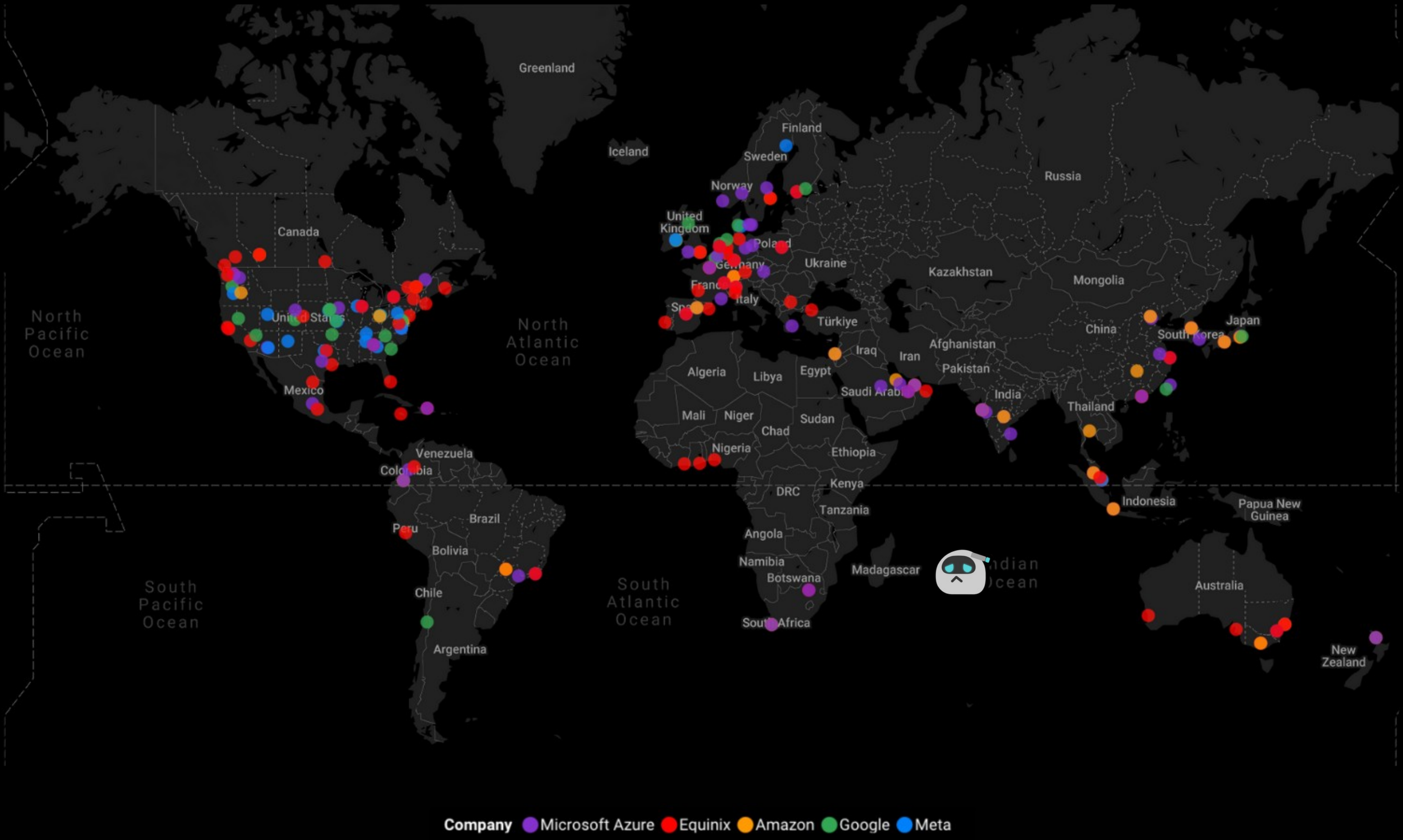
alex.yaml

```yaml
---
apiVersion: v1
kind: Person
metadata:
 name: Alex Bissessur
spec:
  work:
    company: La Sentinelle
    role: Kubernetes Person
    location: Mauritius
  contact:
    website: alexbissessur.dev
    mastodon: moris.social/@AlexB
    github: github.com/xelab04
  interests:
    - Kubernetes
    - Linux
    - Free & Open Source Software
  hobbies:
    - Playing kubectl with Homelab
```

“I do fun things with Kubernetes.”



Data Center Locations (Microsoft, Equinix, Amazon, Google, Meta)



What About Local CSPs?

- Old School – no managed databases, Kubernetes, Load Balancers; VPS only
- Incredibly expensive – cheaper to invest in your own on-prem infra
- Still rely heavily on humans – it is impossible for us to reach the scale and breadth of US hyperscalers
- Cloud Native is completely ignored

Connectivity Crisis

- Bandwidth is the most expensive part of hosting
- Mauritius has mainly only the SAFE undersea cable
- Problem worsened when undersea cables get damaged, making EU/US timeout
- The bandwidth needed – symmetric 1Gb doesn’t exist

Why Am I Complaining?

- A lack of local infrastructure means dependency on foreign systems and companies
- Impossibility for digital sovereignty
- Owning our data and our infrastructure is important given geopolitical tensions
- Handicapping local companies & local talent

A Quick Detour on Sovereign Infra

- NextCloud is not a full replacement of Google/MS clouds
- OVH/Hetzner are not full replacements for American hyperscalers
- Airbus is a full replacement for Boeing (and it wasn’t easy)
- A replacement is only possible when all demands (even the niche ones) are met

My DevCon 24 Workshop

- I ran a workshop on Kubernetes at Devcon ‘24
- Attendees participated on laptops, tablets, phones…
- OS varied from Windows, MacOS, Linux, Android, iOS etc…
- Workshops need consistent environments
  (or you spend more time troubleshooting)
- The cloud was needed, but at what cost?

Running a “Cloud”

- Proxmox – easy & free way to run VMs
- Cloud VMs – are VMs, but they’re expensive
- Shared Kubernetes cluster – not as secure or isolated, and needs a lot of config for multi-tenancy
- HobbyFarm – complete lab platform, FOSS, self-hostable, but relies on AWS/DO

How do we build a cloud?

What is a “Cloud”

- Isolation – different users must stay separate and unable to interact
- Cloud – it must run on someone else’s computer
- Automated – everything must run without human intervention so it is easily scaled

VPS or Containers?

- VMs are heavy – need hardware emulation, runs its own kernel & OS
- Containers – scale very fast
- VMs – more secure, better isolation
- Containers – can share networking, storage
- Containers – cloud-native first

Introducing Virtual Clusters

- KinD – run Kubernetes in Docker, perfect for testing
- Virtual clusters can be deployed to an existing K8s cluster – Kubernetes in Kubernetes
- Existing options:
  - VCluster, Loft Labs
  - K3k, Rancher/SUSE

K3k and VCluster

- Provision virtual clusters on a host
- Virtual cluster lives in pods, with the API server exposed
- Option to passthrough host resources, such as: storageclass, ingressclass, configmaps, secrets
- Virtual clusters are isolated across different namespaces

Other Cloud Native Tech

- Longhorn – distributed, low-resources storage for replicated storage across servers
- Traefik – ingress class and gateway API, serves as a “reverse proxy” to the workloads
- Cert-Manager – automatically provisions TLS certs, in my case from Let’s Encrypt
- K3s – lightweight Kubernetes distribution

KRaft

- A combination of Rust and static HTML
- Backend handles authentication, cluster creation, workspaces, and the entire lifecycle
- Frontend is pretty UI for people to clicky clicky
- Rust side is built on actix web
- Resource handling to monitor cluster use (just reads the metrics server)

KRaft

```
NAME                                        CPU(cores)   MEMORY(bytes)
kraft-auth-7c4646b85c-m76jf                 1m           5Mi
# written in Rust
kraft-cluster-manage-6cf9d74c9b-hnrmv       1m           9Mi
# also written in Rust
kraft-db-8566c5b599-4mtrb                   1m           176Mi
# simple MariaDB pod
kraft-frontend-68d8c7698c-z5nck             0m           7Mi
# nginx with static files
kraft-resource-manage-7d7d979fc4-5qlwz      1m           290Mi
# python container, Flask
```

k3k-rs

- Kube-rs is the crate for interacting with Kubernetes resources from Rust

- Using K3k CRDs properly meant implementing them as Rust structs

- K3k-rs is an offshoot project
  - CRDs reimplemented
  - Convenience functions to manage K3k-related resources

List k3k clusters

```rust
let client = Client::try_default().await?;
let list: Vec<cluster::Cluster> = cluster::list::namespaced(&client, "k3k-namespace").await?;
```

```rust
let client = Client::try_default().await?;

let cluster_schema = k3k_rs::cluster::Cluster {
    metadata: kube::core::ObjectMeta {
        name: Some("test-cluster".to_string()),
        namespace: Some("k3k-namespace".to_string()),
        ..Default::default()
    },
    spec: ClusterSpec {
        // servers: 1, (default)
        // agents: 0, (default)
        // mode: "shared".to_string(), (default)
        // persistence: Some(PersistenceSpec {
        //     r#type: Some("dynamic".to_string()),
        //     storage_class_name: None,
        //     storage_request_size: Some("1G".to_string()),
        // }),
        expose: Some(ExposeSpec {
            LoadBalancer: Some(ExposeLoadBalancer {
                etcd_port: Some(2379),
                server_port: Some(443),
            }),
            NodePort: None,
            Ingress: None,
        }),
        ..Default::default()
    },
    status: None,
};

cluster::create(&client, namespace, &cluster_schema).await?;
```

You’re writing yaml in Rust, how hard can it be?

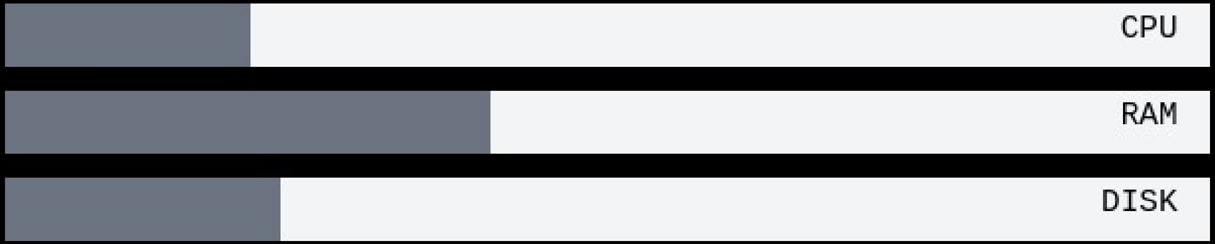
KRaft by Alex

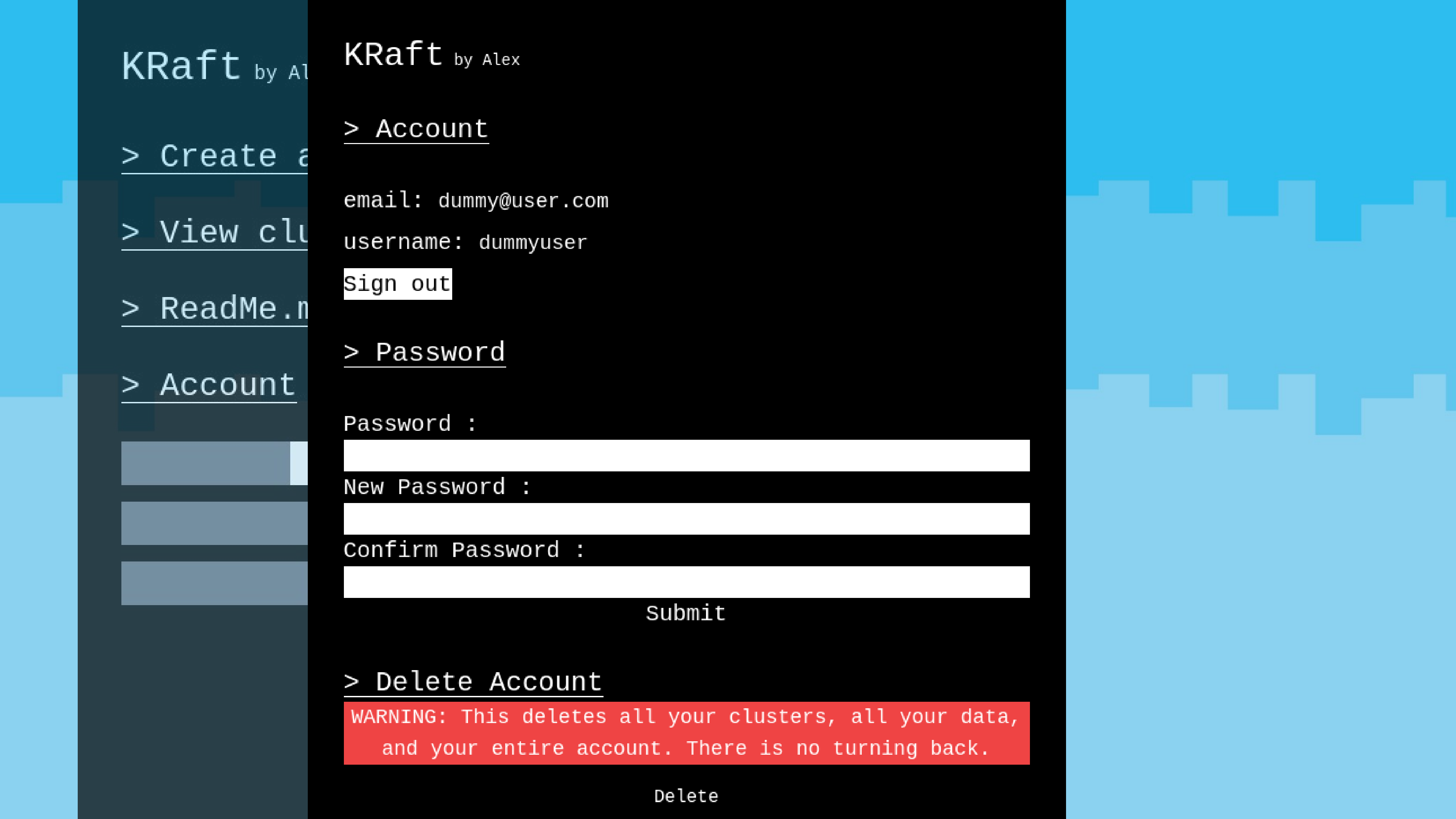
> Create a cluster

> View clusters

> ReadMe.md

> Account




KRaft by Alex
> Account
email: dummy@user.com
username: dummyuser
Sign out
> Password
Password :
New Password :
Confirm Password :
Submit
> Delete Account
WARNING: This deletes all your clusters, all your data, and your entire account. There is no turning back.
Delete

KRaft by Alex

> Create a cluster

Cluster name*:

helpme

TLS-SAN (comma-separated)

helpme.alexbissessur.dev

Submit

KRaft by Alex

> View Clusters

Cluster	Endpoint	Actions
3-test	[redacted]	Expand
3-cnmu		Expand

Cluster Name

3-cnmu

API Server Endpoint

[redacted]

Kubeconfig

Download

KRaft by Alex

> View Clusters

Cluster	Endpoint
3-test	[redacted]
3-cnmu	

Cluster Name

3-cnmu

API Server Endpoint

[redacted]

Kubeconfig

Download

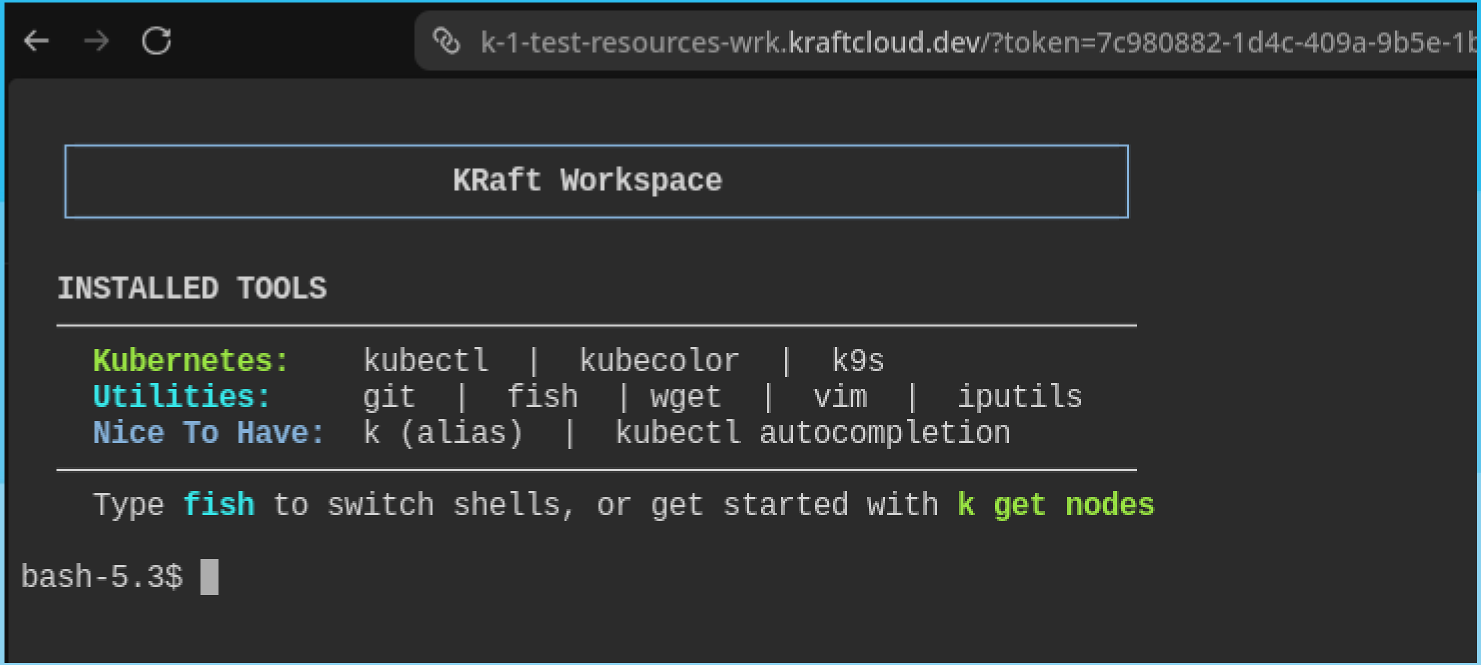
Delete Cluster

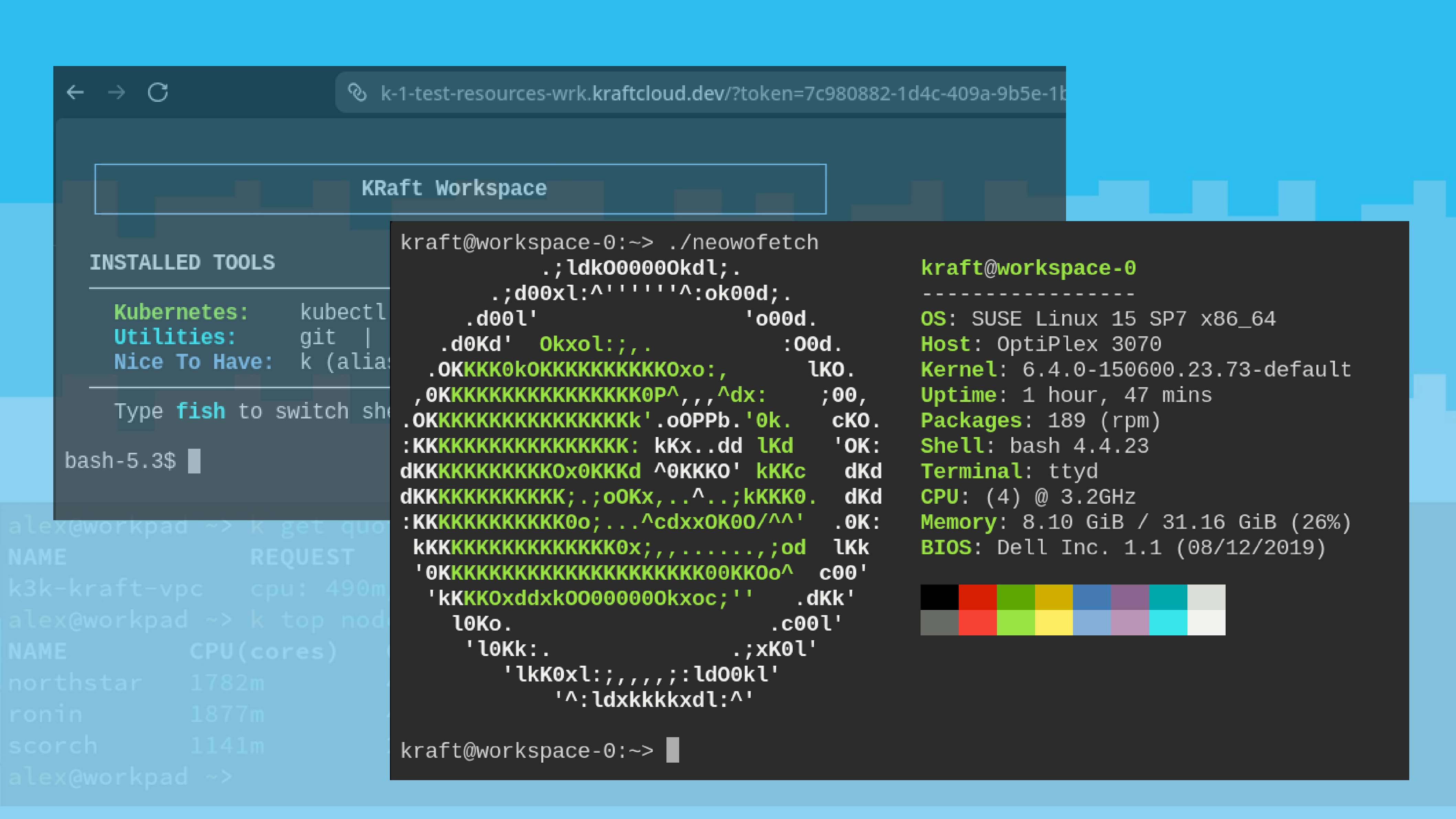
Delete

Cluster Logs

Server   Agents   refresh   copy


```
time="2026-01-05T11:13:51Z" level=info msg="Updated coredns NodeHosts
entry for scorch"
{"level":"info","ts":"2026-01-
05T11:13:54.758942Z","caller":"traceutil/
trace.go:171","msg":"trace[1847929247]
transaction","detail":"{read_only:false; response_revision:8058412;
number_of_response:1;
}","duration":"232.810052ms","start":"2026-01-
05T11:13:54.526114Z","end":"2026-01-05T11:13:54.758924Z","steps":
["trace[1847929247] 'process raft request'  (duration:
232.70677ms)"],"step_count":1}
{"level":"info","ts":"2026-01-
```


k-1-test-resources-wrk.kraftcloud.dev/?token=7c980882-1d4c-409a-9b5e-1b
KRaft Workspace
INSTALLED TOOLS
Kubernetes: kubectl | kubecolor | k9s
Utilities: git | fish | wget | vim | iputils
Nice To Have: k (alias) | kubectl autocompletion
Type fish to switch shells, or get started with k get nodes
bash-5.3$


k-1-test-resources-wrk.kraftcloud.dev/?token=7c980882-1d4c-409a-9b5e-1b
KRaft Workspace
INSTALLED TOOLS
Kubernetes: kubectl
Utilities: git |
Nice To Have: k (alias
Type fish to switch she
bash-5.3$
kraft@workspace-0:~> ./neowofetch
kraft@workspace-0

OS: SUSE Linux 15 SP7 x86_64
Host: OptiPlex 3070
Kernel: 6.4.0-150600.23.73-default
Uptime: 1 hour, 47 mins
Packages: 189 (rpm)
Shell: bash 4.4.23
Terminal: ttyd
CPU: (4) @ 3.2GHz
Memory: 8.10 GiB / 31.16 GiB (26%)
BIOS: Dell Inc. 1.1 (08/12/2019)
kraft@workspace-0:~>

KRaft Admin View

by Alex

[>] View Users

Username	Email	Betacode	
[redacted]		[redacted]	Expand
[redacted]		[redacted]	Expand
[redacted]		[redacted]	Expand
alex		[redacted]	Expand
test		hopeops	Expand


KRaft Admin View
by Alex
[>] View Users
Username
alex
test
KRaft by Alex
[>] View Betacodes
Code
Enabled
cnmu6743
hopeops
testcode
meow
Submit

Keeping Workloads Safe

- Users interact with only their virtual API with isolated etcd
- Clusters are
  - separated by namespace
  - isolated through network policies
  - secured with pod security admission level
- Each cluster has its own CoreDNS

VirtualClusterPolicies

- VCPs provide more control over K3k clusters
- Resource quotas
- Default resource limits/requests
- Allowed Mode
- What resources to sync
- PodSecurityAdmissionLevel
- PriorityClass
- Max resource limits/requests
- Disabling network policy

```
📦[bazzite@dump Downloads]$ export KUBECONFIG=/home/bazzite/Downloads/4-help
📦[bazzite@dump Downloads]$ kubectl get nodes
NAME        STATUS   ROLES   AGE
VERSION
northstar   Ready    agent   100m
v1.33.5-k3s1
ronin       Ready    agent   100m
v1.33.5-k3s1
scorch      Ready    agent   100m
v1.33.5-k3s1
```
22:56

It's aliiiive :o 22:57

WAIT IT WORKS!!!! 22:57

Haha, so surprised xD 22:57

Ah, yes.
HTTP 200 Success
Yet UI shows "An error occurred somewhere between here and the server."
Good start 21:12

😭

Now 0:01

listen 0:01

it might not be covered by a TLS cert 0:01

Buuut 0:01

http://testing.kraftcloud.dev 0:01

I did get a thing to be hosted and it just freaking worked first try 0:01

I am impressed 0:01

(Also it only loads via curl, because you have hsts on that domain) 0:02

Work In Progress

- Waiting on hardware for hosting on Proliant G8
- Helm chart to deploy KRaft easily
- More hosting features & options (eg S3)
- Making my own virtual cluster solution
- Backup and restore methods
- Annoying K3k bugs

Thank You!

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